

ECO2010H1F
Problem Set 5 Solutions

Question 1

(a) An optimal plan satisfies the F.O.C.s

$$u'(c_t) = \beta \lambda_{t+1}, \quad t = 0, 1, \dots, T-1 \quad (1)$$

$$\lambda_t = \beta \lambda_{t+1} F'(k_t), \quad t = 1, \dots, T-1 \quad (2)$$

$$k_{t+1} = F(k_t) - c_t, \quad t = 0, \dots, T-1 \quad (3)$$

$$\lambda_T = v'(k_T) \quad (4)$$

From (3), in any t output can be either consumed or saved. The marginal benefit of additional consumption in period t is $u'(c_t)$. The marginal cost of additional consumption is a reduction in capital available for the subsequent period, $\beta \lambda_{t+1}$. (1) requires that consumption in each t be chosen so that the marginal benefit of additional consumption equals its marginal cost.

(b) At $t+1$, (1) implies that

$$u'(c_{t+1}) = \beta \lambda_{t+2} \quad (5)$$

and (2) in turn implies that

$$\lambda_{t+1} = \beta \lambda_{t+2} F'(k_{t+1}) \quad (6)$$

substituting (5) into (6) yields

$$\lambda_{t+1} = u'(c_{t+1}) F'(k_{t+1})$$

which substituted into (1) results in the Euler equation

$$u'(c_t) = \beta u'(c_{t+1}) F'(k_{t+1}) \quad (7)$$

determining relative consumption between successive periods. The left-hand side is the marginal benefit of consumption in period t , while the right-hand side is the marginal cost, where the marginal cost is measured by marginal utility of potential consumption foregone, $u'(c_{t+1}) F'(k_{t+1})$, discounted one period. The actual level of the optimal consumption path $c_0, c_1, \dots, c_{T-1}$ is determined by the initial capital k_0 and by the requirement (4) that the shadow price of capital in the final period λ_T be equal to the marginal value of the terminal stock $v'(k_T)$.

(c) The Euler equation (7) can be rearranged to yield

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} = F'(k_{t+1})$$

The left-hand side of this equation is the intertemporal marginal rate of substitution in consumption, while the right-hand side is the marginal rate of transformation in production, the

rate of which additional capital can be transformed into additional output. Subtracting $u'(c_{t+1})$ from both sides, the Euler equation (7) can be expressed as

$$u'(c_t) - u'(c_{t+1}) = (\beta F'(k_{t+1}) - 1) u'(c_{t+1})$$

Since u is assumed to be concave, this implies

$$\begin{array}{ccc} > & & > \\ c_{t+1} = c_t & \Longleftrightarrow & \beta F'(k_{t+1}) = 1 \\ < & & < \end{array}$$

Whether consumption is increasing or decreasing under the optimal plan depends on the balance between technology and the rate of time preference.

(d) The Hamiltonian for this problem is

$$H(c_t, k_t, \lambda_{t+1}) = u(c_t) + \beta \lambda_{t+1} (F(k_t) - c_t)$$

which immediately yields the optimality conditions

$$\begin{aligned} \frac{\partial}{\partial c} H &= u'(c_t) - \beta \lambda_{t+1} = 0, \quad t = 0, 1, \dots, T-1 \\ \frac{\partial}{\partial k} H &= \beta \lambda_{t+1} F'(k_t), \quad t = 1, 2, \dots, T-1 \\ \frac{\partial}{\partial \beta \lambda_{t+1}} H &= F(k_t) - c_t = k_{t+1}, \quad t = 0, 1, \dots, T-1 \\ \lambda_T &= v'(k_T) \end{aligned}$$

(e) The Bellman equation is

$$\begin{aligned} v_t(k_t) &= \max_{c_t} \{u(c_t) + \beta v_{t+1}(k_{t+1})\} \\ &= \max_{c_t} \{u(c_t) + \beta v_{t+1}(F(k_t) - c_t)\} \end{aligned}$$

The F.O.C. is

$$u'(c_t) - \beta v'_{t+1}(F(k_t) - c_t) = 0 \quad (8)$$

The value function at $t+1$ is

$$v_{t+1}(k_{t+1}) = \max_{c_{t+1}} \{u(c_{t+1}) + \beta v_{t+2}(F(k_{t+1}) - c_{t+1})\} \quad (9)$$

By the Envelope Theorem,

$$v'_{t+1}(k_{t+1}) = \beta v'_{t+2}(F(k_{t+1}) - c_{t+1}) F'(k_{t+1}) \quad (10)$$

The F.O.C. for (9) is

$$u'(c_{t+1}) - \beta v'_{t+2}(F(k_{t+1}) - c_{t+1}) = 0 \quad (11)$$

Substituting from (11) into (10) yields

$$v'_{t+1}(k_{t+1}) = u'(c_{t+1})F'(k_{t+1}) \quad (12)$$

Substituting from (12) into (8) and using (3) yields the Euler equation (7),

$$u'(c_t) = \beta u'(c_{t+1})F'(k_{t+1}).$$